Department of Physics

PCS 125 – Waves and Fields

Please insert in the circle the initial of your last name

Please Print Clearly:

Student:_____ _____

Last Name *First Name*

Student Number: _____ **Student Section:** _____

(check the bubble sheet)

Final Exam – April 26, 2010

Check the name of your professor:

Dr. Beauchemin		Dr. Carvalho		Dr. Hu		Dr. Tavakkoli		Dr. Yuan	

Read the following attentively

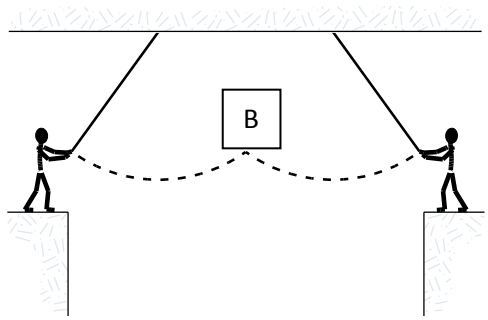
- **Duration of the exam: 3 hours.**
- **The exam consists of 27 multiple-choice questions (1 mark each). Answer ALL questions.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **The work shown in your exam booklet may be checked, <u>at least for one</u> of the questions, selected randomly.**
  - **<u>Transfer your choices to the bubble sheet only when you're sure of them.</u>**
  - **Some answers may be rounded off.**
- **Pagers and cell phones must be silenced and placed out of reach. Hats and earphones are not allowed. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pens, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **Formula sheet provided with the test booklet. No other sheets allowed.**
- **The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) <u>it will be confiscated</u> for the duration of the test.**
- **The exam booklet has a total of 17 pages plus a formula sheet. You are responsible for checking that you have all pages. You can detach the formula sheet.**

_____

Your signature here to signify you read the instructions
(no signature, no mark!)

1.

Two circus clowns (each having a mass of 50 kg) swing on two flying trapezes (of negligible mass and of length 25 m) as shown in the figure. At B, the peak of the swing, one grabs the other, and the two swing back to one platform. The time for the forward and return motion is



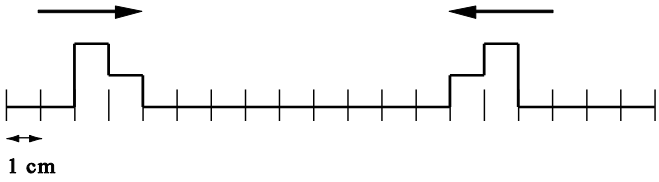
a. 10 s
b. 50 s
c. 15 s
d. 20 s
e. 25 s

2.

Three pendulums with strings of the same length and bobs of the same mass are pulled out to angles θ_1, θ_2 and θ_3 respectively and released. The approximation $\sin\theta = \theta$ holds for all three angles, with $\theta_3 > \theta_2 > \theta_1$. How do the angular frequencies of the three pendulums compare?

a. $\omega_3 > \omega_2 > \omega_1$
b. Need to know amplitudes to answer this question.
c. Need to know $\sqrt{g/L}$ to answer this question.
d. $\omega_1 > \omega_2 > \omega_3$
e. $\omega_1 = \omega_2 = \omega_3$

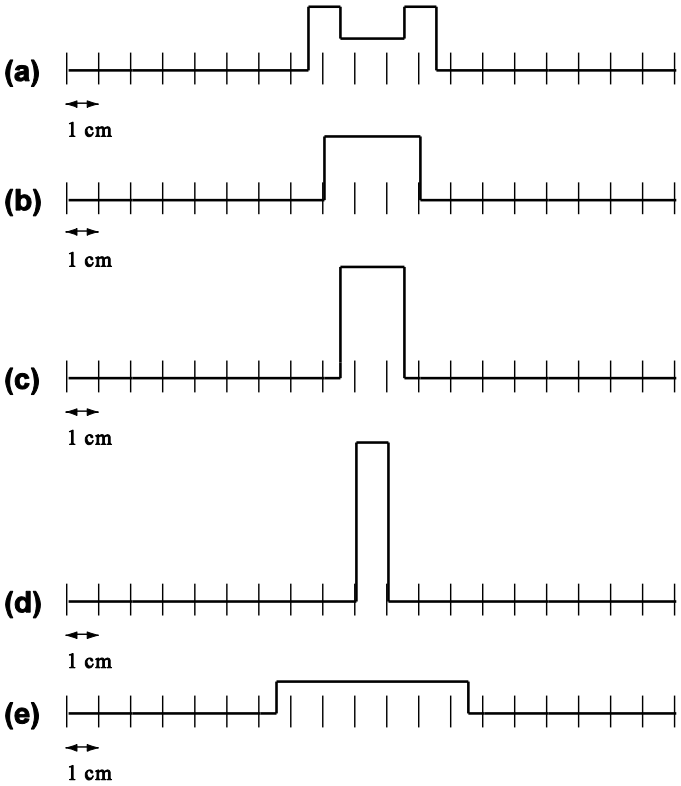
3.



Two pulses travel on a long string towards each other with a speed of 10 cm/s. At t = 0 s their positions are shown in the figure above. Which of the diagrams below correctly illustrates the shape of the string at 0.5 s?

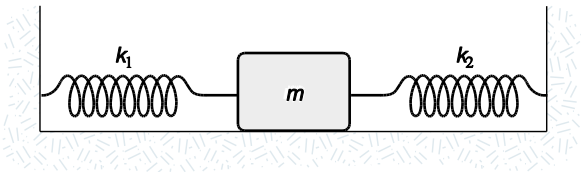
a. (a) b. (b) c. (c)

d. (d) e. (e)



4.

The object of mass m, in the figure, slides on a frictionless surface. If m = 2 kg, k_1 = 800 N/m and k_2 = 500 N/m, what is the frequency of the oscillation (in Hz) when the object is displaced from its equilibrium position?



a. 6
b. 2
c. 4
d. 8
e. 10

5.

The tension in a stretched rubber cord, of mass 12.0 g and length 2.0 m, is 54 mN. What is the speed of a transverse wave in this cord?

a. 2.1 m/s
b. 3.0 m/s
c. 4.5 m/s
d. 6.3 m/s
e. 9.0 m/s

6.

Two harmonic waves traveling along a rope in opposite directions interfere to produce a standing wave described by $y = 3 \sin(2.00\, x) \cos(5.00\, t)$, where x and y are in meters and t is in seconds. What is the transverse speed of the rope at $t = 1.00$ s and $x = 2.00$ m?

a. 2.50 m/s b. 0.40 m/s c. 10.9 m/s d. 2.18 m/s e. 1.11 m/s

7.

One of the strings of my guitar has a fundamental frequency of 440 Hz and another has a fundamental frequency of 660 Hz. Which of the following set of frequencies could be produced on both of these strings?

a. 440 Hz, 880 Hz, and 1320 Hz
b. 660 Hz, 1320 Hz, and 1980 Hz
c. 880 Hz, 1320 Hz, and 1760 Hz
d. 1320 Hz, 2640 Hz, and 3960 Hz
e. None of the above.

8.

A 50-kg satellite circles planet Nikto at an altitude of 1.7×10^{6} m. What minimum energy is required to move the satellite to an outer circular orbit at an altitude of 4.2×10^{6} m?
(Nikto: radius = 6.4×10^{6} m, mass = 6.0×10^{24} kg)

a. 2.9×10^{8} J
b. 3.5×10^{8} J
c. 4.1×10^{8} J
d. 4.7×10^{8} J
e. 5.9×10^{8} J

9.

Three 5.0 kg masses are located at points (0.40, 0) m, (0, 0) m and (0, 0.30) m on the *xy* plane. What is the magnitude of the resultant force on the mass at $x = 0$, $y = 0.30$ m as caused by the other two masses?

a. 2.6×10^{-8} N
b. 2.0×10^{-8} N
c. 2.9×10^{-8} N
d. 2.3×10^{-8} N
e. 2.1×10^{-8} N

10.

A planetary system consists of five planets all of different mass. How many terms appear in the expression for the total gravitational potential energy of this planetary system?

a. 4
b. 5
c. 10
d. 20
e. 25

11.

Two spheres of mass M are separated by a distance R. They are isolated in space with no other masses nearby. The magnitude of the gravitational force experienced by each mass is F. If, instead, the spheres had masses $2M$ and $3M$ and their separation was $2R$, what would be the magnitude of the gravitational force on the larger mass, as expressed in terms of F?

a. $3F$ b. $2F$ c. $F/2$ d. $2F/3$ e. $3F/2$

12.

A laser light beam of wavelength 588 nm in vacuum is incident from air onto a block of unknown material. The incident angle is 45° and the angle of refraction is 30°.
Find the frequency of the light in the material.

a. 2.1×10^{14} Hz
b. 4.2×10^{14} Hz
c. 7.2×10^{14} Hz
d. 3.6×10^{14} Hz
e. 5.1×10^{14} Hz

13.

A small light fixture on the bottom of a swimming pool is 2.50 m below the surface. The light emerging from the still water (n = 1.33) forms a circle on the water surface. What is the diameter of this circle?

a. 5.7 m
b. 2.8 m
c. 2.2 m
d. 4.4 m
e. 3.6 m

14.

Which of the following will cause the fringes in a two-slit interference pattern to get closer together?

a. increasing the wavelength of the light
b. increasing the distance between the slits and the viewing screen
c. decreasing the slit spacing
d. immersing the entire apparatus in water
e. none of the above

15.

Two sources of light illuminate a double slit simultaneously. One emits light of wavelength 550 nm and the wavelength of the other is unknown. The seventh (m = 7) bright fringe of the unknown wavelength overlaps the sixth (m = 6) bright fringe of the light of 550 nm wavelength. What is the unknown wavelength?

a. 471 nm
b. 367 nm
c. 642 nm
d. 413 nm
e. 530 nm

16.

A material having an index of refraction of 1.75 is used as an antireflective coating on a piece of glass whose index of refraction $n = 1.50$. What should the minimum thickness of this film be to minimize reflection of 500 nm light?

a. 1.0×10^{-7} m
b. 2.0×10^{-7} m
c. 7.1×10^{-8} m
d. 1.7×10^{-7} m
e. 1.4×10^{-7} m

17.

A particle with a charge of +40 μC is located on the x axis at $x = -20$ cm, and a second particle with a charge of –50 μC is placed on the x axis at $x = +30$ cm. What is the magnitude of the total electrostatic force on a third particle with a charge of –4 μC placed at the origin ($x = 0$)?

a. 41 N b. 16 N c. 56 N d. 35 N e. 72 N

18.

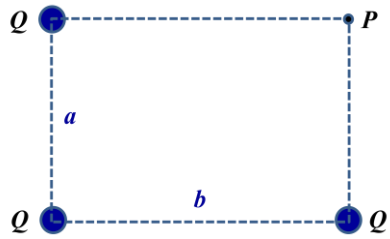
A particle of mass 50 g and charge 5.0 μC is released from rest when it is 50 cm from a second particle with a charge of –20 μC. Determine the magnitude of the initial acceleration of the 50 g particle.

a. 54 m/s^2 b. 72 m/s^2 c. 90 m/s^2 d. 65 m/s^2 e. 36 m/s^2

19.

If Q = +16 nC, a = 3 m, and b = 4 m, what is the magnitude of the electric field at point P?

a. 33 N/C
b. 31 N/C
c. 13 N/C
d. 19 N/C
e. 24 N/C



20.

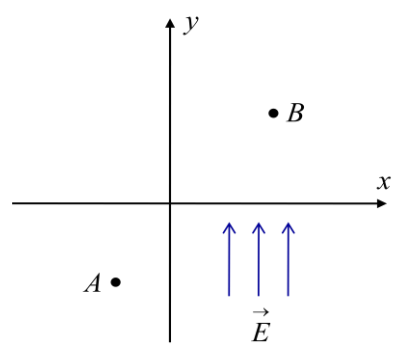
When a negative charge is released and moves along an electric field line, it moves to a position of

a. lower electric potential and higher electric potential energy
b. lower electric potential and lower electric potential energy
c. higher electric potential and lower electric potential energy
d. higher electric potential and higher electric potential energy
e. greater magnitude of the electric field

21.

A uniform electric field of magnitude 200 V/m is directed in the positive *y* direction as shown in the figure. The coordinates of points *A* and *B* are *A*(-2,-3) m and *B*(4,4) m.
What is the electric potential difference $V_B - V_A$?

a. -1.4 kV
b. +1.4 kV
c. -1.2 kV
d. +1.2 kV
e. None of the above



22.

In a particular cathode-ray tube the beam current is measured to be 50 μA. How many electrons on average will strike the screen of the tube screen during an interval of time of 30 seconds?

a. 3.5×10^{14}
b. 8.4×10^{14}
c. 4.7×10^{15}
d. 9.4×10^{15}
e. Impossible to calculate based on the given information.

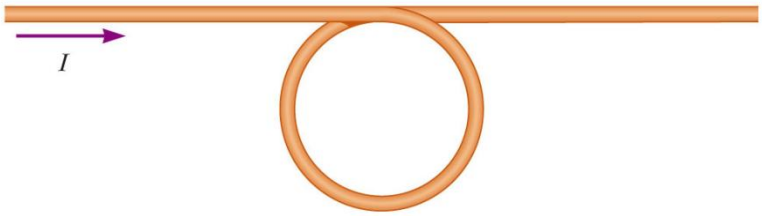
23.

An electron travels with no acceleration through a region of space where an electric field **E** and a magnetic field **B** exist.
Which one of the following statements is the best conclusion?

a. Both **E** and **B** must be zero in that region
b. **E** must be zero, but **B** might be non-zero in that region
c. **E** and **B** might both be non-zero, but they must be mutually perpendicular
d. **B** must be zero, but **E** might be non-zero in that region
e. **E** and **B** might both be non-zero, but they must point in opposite directions

24.

A conductor consists of a circular loop (with a diameter of 2.0 cm) and two straight, long sections as shown in the figure.
The wire lies in the plane of the paper and carries a current of 5.0 A.
What is the strength and direction of the magnetic field at the center of the loop?

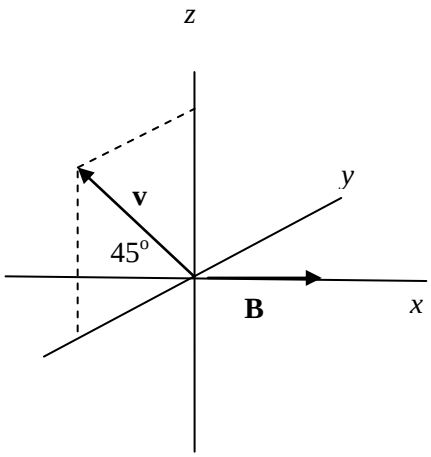


a. 2.1×10^{-4}T (out of the page)
b. 2.1×10^{-4}T (into the page)
c. 4.1×10^{-4}T (out of the page)
d. 4.1×10^{-4}T (into the page)
e. 4.1×10^{-4}T (upward)

25.

An electron moves in the *yz*-plane with a speed of 10^7 m/s at an angle of 45°, as shown in the figure. In the region where the electron travels there is a magnetic field **B**= 0.50 **i** T.
What is the magnetic force exerted on the electron?

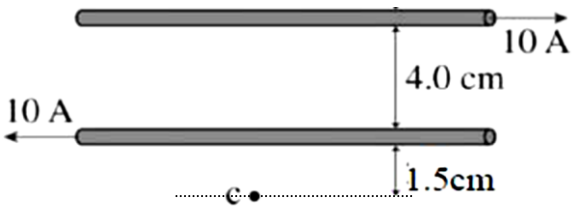
a. 5.7×10^{-13} (–**j** – **k**) N
b. 5.7×10^{-13} (**j** – **k**) N
c. 5.7×10^{-13} (**j** + **k**) N
d. 5.7×10^{-13} (–**j** + **k**) N
e. 5.7×10^{-13} (**j** + **i**) N



26.

Two parallel wires each carry, in opposite directions, a 10 A current. What is the magnitude and direction of the magnetic field at point **c**?

a. 1.7×10^{-4} T (out of the page)
b. 1.7×10^{-4} T (into the page)
c. 9.7×10^{-5} T (downward)
d. 9.7×10^{-5} T (into the page)
e. 9.7×10^{-5} T (out of the page)



27.

Two long parallel conductors, separated by 10 cm, carry currents in the same direction. The first wire carries a current I_1 = 5 A and the second carries a current I_2 = 8 A. What is the magnitude of the force per unit length exerted by I_1 on I_2?

a. 10^{-5} N/m b. 5×10^{-5} N/m c. 8×10^{-5} N/m d. 5×10^{-7} N/m e. 8×10^{-7} N/m